Summary Report: Big Data Analytics – TP02

Student: Mamma Sai

Executive Summary

This study benchmarks three strategies for processing a **5.31 GB** CSV dataset containing **55.4 million rows** on a mid-range system (**15.3 GB RAM**, **12 CPU logical cores**).

Evaluated Approaches

1. **Standard Pandas Loading (`pandas.read_csv`):** Failed due to an **Out-Of-Memory (OOM)** error. System RAM usage exceeded 92%, triggering a safety guard to terminate the execution before system crash.
2. **Stream Processing (Pandas Chunking):** Successfully processed the dataset sequentially in chunks of 100,000 rows. It maintained a very low, near-constant memory footprint (**142 MB**), though requiring more time (**128.5 s**).
3. **Parallel Out-of-Core Compute (Dask DataFrame):** Successfully completed in **35.5 s** ($\approx 3.6\times$ faster than chunking) by leveraging multi-threading and lazy evaluation (*Task Graphs*). Peak memory reached **2155 MB**.

Performance Benchmark

Method	Status	Execution Time (s)	Peak RAM (MB)
Standard Pandas	OOM (Aborted)	—	3935 (Exceeded 92%)
Pandas Chunking	Success	128.5	**142**
Dask DataFrame	Success	**35.5**	2155

Key Takeaways

- **Pandas Chunking** is ideal for severely memory-constrained environments where low resource usage is mandatory.
- **Dask** is preferred when speed and multi-core scalability are the main priorities, using moderate memory in exchange for significantly faster compute times.